The Iranian Rial as an Investment

Valuation, Relief Scenarios, and the Executability of an Offshore Position

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Abstract. This paper evaluates a specific trade: a foreign retail investor buys Iranian rials (IRR) at the October 2026 Tehran free-market rate of about 2.69 million per US dollar and holds them against the possibility of sanctions relief. The pitch rests on two true facts. The rial trades at about 1.2 percent of its JCPOA-era dollar value and at roughly one tenth of its purchasing-power-parity rate. We ask three questions in order. Is the currency undervalued? What does relief do to currencies like this one, on the record? And can an offshore holder acquire, hold, and above all exit the position? The answers are: yes, conditionally; far less than the pitch assumes, because in all seven unification episodes in the comparison set the exchange-rate gap closed by official-rate devaluation and in none by market-rate appreciation; and no, not at any documented price, because the documented Western bid for the rial is USD 0.00 and the lawful export quantity of notes from Iran is IRR 500,000, worth about USD 0.19. A five-scenario model built from the precedent record gives a best single outcome of 1.175× (partial relief, one year, frictionless) and probability-weighted theoretical returns of −7.8, −10.6, −37.6, −57.9, and −83.5 percent at six months, one, three, five, and ten years: the trade fails on expected value before any friction is applied. Under the documented exit bid, the realisable recovery on a USD 10,000 stake at one year is USD 0.07 to USD 0.22. No legal instrument available to this investor pays off if and only if the rial appreciates.

Keywords: Iranian rial; sanctions; exchange-rate unification; capital controls; sanctions relief; currency speculation; executability.

JEL codes: F31, F38, F51, G15.

Non-technical summary. The Iranian rial has collapsed. One dollar bought about 32,000 rials when the 2015 nuclear deal was signed; in October 2026 it bought about 2.69 million. That collapse tempts a familiar idea: buy the currency cheaply, wait for a political settlement, and sell into the recovery. This paper tests that idea three ways. First, is the rial actually cheap? By the usual yardsticks, yes: prices in Iran are about a tenth of US prices when converted at the market rate, which is an extreme gap. Second, when countries in this position settle with their adversaries or unify their exchange rates, does the market rate jump? The historical record, covering Iran itself in 2013–2016 and six other countries, says no. Relief rallies were worth 3 to 20 percent and faded within months; every exchange-rate unification on record closed the gap by devaluing the official rate down to the market rate, never by the market rate rising. Third, and decisively, can a foreigner actually do the trade? Buying is possible only informally or in trivial amounts; deposits pay interest far below inflation; and selling is the binding failure: no Western dealer bids for the currency at all (the documented buyback price is zero), Iranian law allows a traveller to carry out only about 19 US cents' worth of rial notes, and US law since 2013 has been designed to make the rial unusable outside Iran by sanctioning any bank that intermediates it. Our scenario model finds that even a hypothetical holder who could trade freely, at no cost, at Tehran prices, loses money on average in every period examined, because continued depreciation in the most likely scenario outweighs the bounded gains in the relief scenarios. The recommendation embedded in the evidence is not "the rial is expensive." It is that this particular asset, for this particular holder, is not a tradeable asset.

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1 Introduction

The trade is usually pitched in one sentence. The Iranian rial has lost about 98.8 percent of its dollar value since the 2015 nuclear agreement (Reconstructed, A), a war and a naval blockade have

driven Iran's oil exports to zero, and if any settlement restores even part of the pre-war economy, a currency priced at one tenth of purchasing power parity must appreciate several-fold. Versions of this pitch attach to every deeply depreciated sanctioned currency. It deserves a precise answer rather than a dismissive one, because its first premise is correct: the rial is, on price-level evidence, one of the cheapest currencies in the world (Section 2.4). The question is what follows from that.

Three distinct claims are bundled inside the pitch, and they fail or survive independently.

1. **Value.** The free-market rate of 2,674,000 rials per dollar (OBSERVED, B) sits far below any purchasing-power benchmark (Section 2.4 puts the gap at about 10.5–11×) and far below the official-complex rate of about 1.75–1.77 million at which the state actually transacts (Section 2.1). A gap of this size is a fact. Whether it is an opportunity depends on the next two claims.

2. **Dynamics.** The pitch assumes that political relief converts into nominal appreciation of the rate a speculator can buy. The precedent record (Section 4) is the evidence on this conversion, and it is unfavourable: relief rallies in Iran's own 2013–2016 episode were worth at most about 20 percent cumulatively and were fully retraced by drift; and in the seven exchange-rate unification episodes in the comparison set, the official–market gap closed by official devaluation in seven of seven cases and by market appreciation in zero of seven.

3. **Executability.** The pitch assumes a position can be established and liquidated at something near the quoted rate. For an offshore retail holder this is false in a way that can be documented price by price: the only Western buyback quote located for the rial is USD 0.00 (OBSERVED, B), and Iranian law caps the rial notes a traveller may carry out at IRR 500,000 (OBSERVED, A), about USD 0.19 at the entry rate used in this paper.

The paper answers the three questions in that order, then prices the trade. Section 2 documents the market as it stands in October 2026: the rate complex, the 2025–2026 collapse, the macro state (dual official inflation prints of 73.6 and 68.4 percent (OBSERVED, B), money growth, the IMF output path, fiscal dominance), the war baseline, reserves and frozen assets, and the enacted redenomination. Section 3 sizes the relief upside in foreign-exchange terms: incremental oil revenue of USD 26.5, 51.5, and 79.5 billion per year in low, base, and high cases against the current zero-export baseline (MODELLED, B). Section 4 assembles the precedent record on what relief and unification do to market exchange rates. Section 5 builds a five-scenario model from that record and computes theoretical returns for stakes of USD 1,000, 10,000, and 100,000 under zero frictions. Section 6 documents executability: the legal bars, the acquisition channels, the holding regime, and the exit. Section 7 applies the exit evidence to the model and computes realisable returns. Section 8 states what 2×, 5×, and 10× outcomes would require and tests those requirements against the precedent record. Section 9 examines the alternative exposures. Section 10 gives the verdict and the conditions that would change it. Section 11 states limitations. Section 12 concludes.

The preview is as follows. The best single cell in the scenario model is partial relief at one year, 1.175× before frictions (MODELLED, B). Probability-weighted theoretical returns are negative at every horizon, reaching −83.5 percent at ten years. Realisable returns at the documented bid are −100.0 percent at every horizon and every stake, to one decimal place. The few realisable cases that approach break-even require an in-person exit in Tehran, stakes small enough to fit inside a capped deposit account, and exit channels that the evidence does not document at size.

This is a retrospective analysis of a hypothetical position; there was no pre-registered plan, and no position was taken. The investor profile assumed throughout is a Canadian resident retail investor with US financial-system touchpoints (US-dollar accounts, US correspondent banking somewhere in

the payment chain, possible US travel and brokerage relationships). The legal analysis is research, not legal advice.

2 The market as it stands

2.1 The rate complex

Iran does not have an exchange rate; it has a rate complex, and which rate a figure uses changes its meaning by orders of magnitude. Table 1 gives the complex as of mid-October 2026.

Table 1: The Iranian rate complex, October 2026 (rials per US dollar)

Rate	Value	Date	Grade
Free market (open market, sell)	2,674,000	16 Oct 2026	Observed, B
Free market (tracker range)	2,688,000–2,695,000	3 Oct 2026	Observed, B
ICE negotiated rate (sell)	1,768,100	6 Oct 2026	Observed, B
CBI SANA reference rate	1,773,407	7 Oct 2026	Observed, B
Free market, euro (per EUR)	3,005,000–3,050,000	7–16 Oct 2026	Observed, B
Legacy "official" rate	42,000 (defunct)	n/a	Observed, A

Notes: ICE is the Central Bank of Iran's Iran Currency Exchange Center, operating since November 2023, where exporters must offer foreign exchange and importers buy it; SANA is the CBI's published average of licensed exchange-office rates. The free market trades about 51 percent above the official complex. Tracker dispersion in fast markets runs 5–10 percent between sources; the mid-October cluster from three independent trackers is 2.67–2.70 million. The 42,000 rate is the April 2018 unification rate; it survives only as a frozen statistical series (the World Bank's official-rate series is fixed at exactly 42,000 for 2019–2023 and ends in 2023) and is not an operative transaction rate in 2026. Citing it as "the official rate" is an error this paper does not make.

Source: WSNext (2026); Reuters (2026c); Iran News Wire (2026); World Bank (2026).

Three features of the complex matter for the trade. First, the operative official rate (the ICE negotiated hall and the SANA reference, converged near 1.75–1.77 million) is itself depreciating at about 4–5 percent per month (Observed, C); the state does not defend a level, it manages a crawl. Second, intervention does not move the free rate at current scale: state banks announced sales of up to USD 2 billion after the record low of 3 October 2026, and the rate stayed at 2.67–2.70 million (Observed, B). Third, the free rate is corroborated independently by the gold-coin market: the Emami coin price of 6 October implies a rate of about 2.70 million, within about 1 percent of the quoted free rate, so the coin market currently carries little bubble premium (Reconstructed, B). The free rate is a real price, formed in a cash market, for those physically present in it. That qualifier is the subject of Section 6.

2.2 The collapse path, 2025–2026

The pre-2025 configuration, a free rate near 1.0–1.3 million and inflation of 40–50 percent, no longer exists. The verified event chain is: February 2025, the United States restores "maximum pressure" and the rial breaks 900,000 (Observed, B); March–April 2025, the rate crosses 1,000,000 for the first time (Observed, B); 28 September 2025, UN snapback sanctions are reimposed with the free rate near 1.10–1.15 million (Observed, B); December 2025, a US-engineered dollar shortage, a bank failure, and a bank run take the rate from about 1.32 million (18 December, Associated Press) to about 1.42 million by late December, with bazaar protests beginning on 28 December (Observed, B); February 2026, war begins (Section 2.5); July 2026, a US naval blockade of Iranian ports is in

force and the rate is 1.70–1.80 million (Observed, C); 29 September 2026, 2,558,000 (Bonbast) (Observed, B); 3 October 2026, a record 2,688,000–2,695,000 (Observed, B); 15–16 October 2026, about 2,674,000–2,687,000 (Observed, B). The twelve months October 2025 to October 2026 alone raised the dollar price by 132 percent (Reconstructed, A). Against the JCPOA anchor, the move is larger: 32,000 in July 2015 to 2,674,000 is an 83.6× rise in the dollar price, so the rial retains about 1.2 percent of its deal-era dollar value (Reconstructed, A). Since September 2011 (about 13,000), the rise is 205.7× (Reconstructed, A).

2.3 The macro state

Table 2 summarises the macro state. Four elements bear directly on the currency.

Table 2: Iran: macro summary, latest observations

Indicator	Value	Period	Grade
Inflation, SCI, 12-month average	73.6%	to Sep 2026	Observed, B
Inflation, SCI, point-to-point	89.8%	Sep 2026	Observed, B
Inflation, SCI, food, point-to-point	121.8%	Sep 2026	Observed, B
Inflation, CBI, 12-month	68.4%	to Sep 2026	Observed, C
Inflation, IMF WEO average CPI	50.9%; 68.9% proj.	2025; 2026	Observed/Modelled, A
Broad money (M2) growth	+36.8% y/y	Sep 2025	Observed, B
Real GDP growth	−1.5%; −6.1% proj.	2025; 2026	Observed/Modelled, A
GDP, current USD (WEO)	$371bn; $300bn proj.	2025; 2026	Observed/Modelled, A
General government balance	−5.7% of GDP	2025	Observed, A
General government gross debt	37.3% of GDP	2025	Observed, A
External debt	$5.4bn	Sep 2025	Observed, A
Crude oil production	1.96 mb/d	Sep 2026	Observed, B
Crude oil loadings	0	Sep 2026	Observed, B
Current account	−1.8% of GDP proj.	2026	Modelled, A

Notes: Iran publishes two official inflation series (SCI and CBI) that persistently disagree; both are reported, neither is suppressed. The IMF country page of July 2026 vintage projects 2026 real GDP at −5.4 percent against −6.1 percent in the library WEO pull; the spread is vintage uncertainty and is flagged rather than resolved. GDP in USD is conversion-sensitive: at the October 2026 free-market rate, the same rial GDP converts to well under USD 1,500 per capita against USD 3,924 in the WDI 2025 figure (Reconstructed, B).

Source: Statistical Center of Iran (2026); Central Bank of Iran (2025); International Monetary Fund (2026b); International Monetary Fund (2026a); OPEC (2026); Reuters (2026a).

Inflation. Twelve-month inflation to September 2026 is 73.6 percent on the Statistical Center of Iran's average measure and 89.8 percent point-to-point, with food at 121.8 percent (Observed, B); the Central Bank of Iran prints lower, 68.4 and 83.8 percent (Observed, C). The two-series disagreement is longstanding and is itself a data-quality fact. On either series, inflation at 69–90 percent mechanically requires continuous nominal depreciation of roughly 5 percent per month to hold any real exchange rate. Claims of true inflation near 143 percent circulate in opposition media; they are unverified and are not used (Assumed, C).

Money and fiscal dominance. Broad money grew 36.8 percent year on year to September 2025 (Observed, B); CBI statements put liquidity growth at 19 percent in the first five months of 1405 (April–August 2026), an annualised pace near 50 percent (Observed, C). The financing chain is documented and structural: a budget deficit of about 6 percent of GDP on the budget-law basis (over 950 trillion tomans for fiscal 2025/26 (Observed, C); −5.7 percent of GDP on the WEO

general-government measure (Observed, A)) is financed through state banks whose recourse reaches the central bank; in December 2025 the chain operated in plain sight, when a bank failure and run ended in central bank money creation. Headline public debt is low (37.3 percent of GDP (Observed, A)) because the burden sits off-budget in arrears, state-bank balance sheets, and pension liabilities, and because domestic rial debt is inflated away. The CBI has no operational independence in the sense a disinflation anchor requires: the governor serves at the President's pleasure, a predecessor was dismissed in December 2022 explicitly over the exchange rate, and the CBI publicly describes the free-market rate as a fear rate rather than a policy target. This assessment is the desk's, argued from (Observed, B) sources. Its implication for the trade is direct: the depreciation engine is fiscal, and no document in the evidence base shows it switched off in any year, including the relief years 2013–2017, when liquidity growth never fell below about 25 percent.

Output. Real GDP fell 1.5 percent in 2025 and is projected to fall 6.1 percent in 2026 (Modelled, A); dollar GDP falls from USD 371 billion to a projected USD 300 billion (Modelled, A). External debt is trivially small at USD 5.4 billion (Observed, A): Iran's constraint is access to foreign exchange, not solvency in the conventional external sense.

Reserves and frozen assets. Iran publishes no usable reserves total; the total is Unknown and any paper that prints one is inventing it. The accessible figure in circulation, about USD 24 billion, is unattributed (Assumed, C). Iranian official claims of inaccessible assets exceed USD 100 billion; the April 2026 location breakdown (Appendix Table 11) sums to roughly USD 44 billion across China (about USD 20 billion, of which President Pezeshkian said in September 2026 that "not a single cent" is withdrawable), India, Iraq, the Korea-to-Qatar escrow, the United States, Japan, and the EU (Assumed, C). Independent analyses put the realistically releasable subset at USD 30–35 billion, much of it pre-committed (Assumed, C). The consequence for scenario design is that a settlement's stock effect is bounded at roughly USD 25–40 billion of incremental liquid foreign exchange; its flow effect, restored oil sales, is larger and is priced in Section 3. A related data-quality fact: Iran's last IMF Article IV consultation was in March 2018 (Observed, A).

2.4 Valuation: cheap, and priced that way for identifiable reasons

The World Bank's PPP conversion factor for Iran is 147,938 rials per international dollar for 2025 (Observed, A). Rolled forward with 2026 inflation (IMF average CPI of 68.9 percent for Iran against about 3 percent for the United States) it reaches about 240,000–255,000 (Modelled, B). The free-market rate is therefore about 10.5–11× the PPP rate, and Iran's implied price level is about 9 percent of the US level at the free rate. The World Bank's real exchange rate index (2015 = 100) stood at 78.0 in its 2024 forecast vintage, about 22 percent below 2015 after the 2018–2020 real collapse (World Bank, 2024) (Observed, A); with 2025–2026 nominal depreciation running ahead of even 69–90 percent inflation, the real rate has fallen further since (direction (Observed, B), magnitude (Modelled, B)). No BIS effective exchange rate exists for Iran (the BIS set covers 64 economies and Iran is not among them (Observed, A)), and no IMF effective-rate series was retrievable; the valuation statement rests on PPP and the World Bank index.

The desk's verdict on valuation is two-part. On price levels, the rial is deeply undervalued: about ten times PPP, with a price level under a tenth of the US level. But the discount is not a pricing error that arbitrage can close; it is a price on identifiable impairments: a blockade that has zeroed the main foreign-exchange earner, fiscal dominance with money growth of 37–70 percent, inflation that requires continuous depreciation, capital controls with a multiple-rate system, and no IMF programme or nominal anchor. Two caveats cut the PPP figure itself: Iran's CPI basket is contaminated by administered prices (energy, bread, medicine) and by 2026 price controls and voucher rationing,

so PPP computed off a controlled basket overstates the currency's fair value for tradables. The undervaluation is real but conditional: it is the value of an option on sanctions relief and monetary stabilisation. Sections 4 and 5 price that option from the precedent record rather than from the PPP gap.

2.5 A war baseline, not a sanctions baseline

Any current number for Iran must state its regime. The baseline for this paper is war. War began on 28 February 2026; the Strait of Hormuz closed; a US naval blockade of Iranian ports, imposed in mid-April 2026, relaxed briefly in a June–July ceasefire and waiver window, and reimposed on 13 July 2026, has held since. Seaborne crude loadings fell from about 2.0 mb/d in March 2026 to 0.74 in July, 0.21–0.26 in August, and zero in September 2026, the first month with no Iranian crude cargo loaded since tracking began in 2013; no loadings at all occurred after 25 August (Observed, B). Crude production fell from a 2025 average of 3.26 mb/d to 1.96 mb/d in September 2026 (Observed, B). About 80 million barrels sit in floating storage (Reconstructed, B). The pre-war baseline was itself sanctions-constrained: crude deliveries of 1.4–1.6 mb/d, about 91 percent to China (Reconstructed, B), at discounts to ICE Brent of USD 2.50–12 per barrel across 2025 to February 2026 (Reconstructed, B), worth USD 43 billion of oil export revenue on the EIA's calendar-2024 estimate (Reconstructed, B) to USD 67 billion on the CBI's broader fiscal-1403 definition (Observed, B). Customs non-oil exports were USD 57.8 billion in fiscal 1403, of which roughly half is hydrocarbon-linked (petrochemicals, LPG, gas, bitumen) (Observed, B), and they fell 28.2 percent year on year in March–July 2026. Foregone oil revenue under the blockade runs at about USD 4–5.5 billion per month against the March 2026 pace (Modelled, C). Total Hormuz crude clearance averaged 11.62 mb/d in the week to 13 October 2026, about 68 percent of the pre-war baseline (Observed, B), with ICE Brent settling at USD 105.92 on 15 October 2026 (Observed, B).

Iran's endowment is not the constraint. Proven oil reserves are 208.6 billion barrels on the OPEC member-reported basis (OPEC, 2025) (Observed, A) and 157.8 billion on the Energy Institute basis (Energy Institute, 2025) (Observed, B); gas reserves are 33,988 billion cubic metres (OPEC) (Observed, A) against 32.1 trillion cubic metres (Energy Institute) (Observed, B): third in the world for oil, second for gas, with reserve life above a century at current output on either basis (Modelled, B). The constraint is that the molecules cannot currently be sold.

2.6 The redenomination: enacted, dated, and adverse to offshore notes

On 5 October 2025 parliament enacted the redenomination: one new rial equals 10,000 current rials, the unit keeps the name "rial", and the subunit is the qeran (Reuters, 2025) (Observed, B). The CBI has up to two years to prepare (to about late 2027), followed by a three-year dual-circulation window in which old and new units coexist; conversion of notes runs through the Iranian banking system. The fiscal 1405 budget bill was already drafted in new rials (Observed, B), and in October 2026 the CBI introduced a new 10,000,000-rial banknote, the largest denomination in its history, worth about USD 3.7 at the free rate (Observed, C). Redenomination is arithmetic, not appreciation: the conversion ratio changes no value by itself. Its relevance to this paper is mechanical and adverse. No provision for exchanging notes held abroad was located in anything retrieved (Assumed, B); an offshore holder must present notes inside Iran within a defined window (plausibly 2027–2030) or hold paper whose only certain market is the collectible trade. Section 7 prices this as a dated hazard with a modelled window close in October 2030. Secondary reporting conflicts on the new unit's name (one outlet

reports "new toman"); the conflict is recorded as an open item and does not affect the mechanics.

3 The relief upside in foreign-exchange terms

Relief matters to the currency only through foreign exchange: how much arrives, how fast, and against which baseline. This section sizes the flow. The model's scenario definitions in Section 5 use these magnitudes as plausibility anchors; no elasticity from oil revenue to the exchange rate is estimated anywhere in this paper, because the evidence base contains none and inventing one would be fabrication.

3.1 How fast volumes can return

Three precedents bound the restart speed. In 2016, from Implementation Day on 16 January, crude exports rose from about 1.1 mb/d to a 2.09 mb/d average within the first year, and production reached about 3.7 mb/d by late 2016: with fields intact and buyers willing, Iran added about 1 mb/d of exports in six to twelve months (Observed, A). In the June–July 2026 ceasefire window, loadings rebounded to about 1.35 mb/d within three weeks before the blockade snapped back (Reconstructed, B): the export machine restarts in weeks when access opens; the constraint is political, not logistical. Wood Mackenzie's restart curve for war-disrupted fields reaches 70 percent of prior output in three months and 90 percent in six (Modelled, C). Against these anchors sit three counterweights: production starts from 1.96 mb/d, not 3.26, and shut-in wells at mature fields do not all return; war damage to export infrastructure is unquantified (Unknown); and the EIA's pre-war capacity assessment of 3.8 mb/d (Reconstructed, B) is an upper reference, not a current measurement. Post-war sustainable capacity is Unknown in the retrieved sources.

3.2 The modelled relief cases

Table 3 gives the modelled export paths at a twelve-month run rate. The price deck is set below the 15 October 2026 Brent settle of USD 105.92, on the stated assumption that relief removes part of the war premium embedded in that print; prices are graded Assumed in the source model, with no quality letter assigned. Volumes are (Modelled, B) from the restart precedents.

Table 3: Relief upside: incremental crude and condensate revenue at a 12-month run rate, against the September 2026 zero-export baseline

Case	Volume (mb/d)	Brent ($/bbl)	Discount ($/bbl)	Net ($/bbl)	Incremental FX ($bn/yr)
Low	1.1	70	4	66	26.5
Base	1.7	85	2	83	51.5
High	2.2	100	1	99	79.5

Notes: Incremental means against the September 2026 outcome of zero crude loadings. The discount narrows from the 2025 average of about $5 for Iranian Light as legal buyers re-enter and intermediation layers fall away. Add-ons, modelled separately to avoid double counting: sale of about 80 million barrels of floating storage, a one-off USD 5–8 billion in the first one to two quarters (Modelled, B); and normalisation of products, petrochemical, and non-oil trade, USD 5–10 billion per year (Modelled, C). Total foreign-exchange swing against the blockade baseline: about USD 32, 60, and 88 billion per year in the low, base, and high cases, plus the storage sale.

Source: desk model documented in the data appendix; volumes anchored to OPEC (2026), U.S. Energy Information Administration (2024), AInvest (2026); pre-war earnings range EPRINC (2024), Hellenic Shipping News (2025); non-oil exports IRNA (2025).

Two qualifications decide how much of this flow a currency holder should capitalise. First, the baseline: against the pre-war configuration (oil revenue of USD 43–67 billion plus USD 57.8 billion of non-oil exports), the base case is restoration, not expansion. Only the high case, volumes above anything achieved since 2018 at near-par pricing, exceeds the status quo ante materially. Relief mostly removes a collapse discount; it does not create a new earnings level. Second, pre-commitment: explicit energy subsidies were USD 41.7 billion in 2020 on the World Bank estimate (World Bank, 2020) (Observed, B), retail gasoline sells at about half a US cent per litre against a distribution cost several times the price, and the gasoline balance runs a deficit of about 28 million litres per day (about 175 kb/d) met by imports (Shana, 2026) (Observed, B): a large share of any windfall is spent holding domestic energy prices down and covering the fuel gap before it can accumulate as reserves. Payment friction further separates gross revenue from usable foreign exchange: China trade settles mainly in yuan through small banks, Iraqi gas receipts sit in a restricted account releasable only with US approval, and escrowed proceeds have historically remained blocked for years after the underlying sale completed. The size of total intermediation leakage is Unknown in the retrieved sources.

3.3 What caps the upside beyond two years

The Seventh Development Plan markets an oil-sector programme requiring about USD 190 billion of financing, explicitly including foreign finance (Observed, B); the national oil company cannot fund it from cash flow at sanctioned prices, and the 2016 precedent is that international oil company money is what moves capacity above about 3.8 mb/d. Mature-field decline is real but unquantified in the retrieved sources (Unknown). On the institutional side, Iran has been exempt from OPEC production cuts since December 2018; for 2027 baselines the OPEC+ capacity assessment by DeGolyer and MacNaughton excludes Iran, Russia, and Venezuela because of US sanctions (Reuters, 2026b) (Observed, B), so relief volumes up to about 2 mb/d face no formal quota barrier, while a return above that level reopens a baseline fight Iran currently cannot even be measured for. Mapping to Section 5: the low case is the plausible scale of partial relief (waivers and tolerated sales without banking reconnection); the base case fits a comprehensive deal; the high case is attainable only under full reintegration.

4 What relief does to a currency: the precedent record

The model's appreciation assumptions come from this section and nowhere else. The evidence is the case record: Iran's own 2011–2020 history, six comparison episodes of unification or normalisation, and two side cases (Russia 2022, Iraq) that isolate the exit and the retail-speculation base rate. Free-market prints in thin markets are point quotes, not fixings; individual prints carry at least ±5 percent uncertainty, worse in panics (Observed, B).

4.1 Iran, 2011–2020: the central precedent

The 2012 shock. Under the EU oil embargo and the March 2012 SWIFT disconnection, the free rate went from about 13,000 in late 2011 to intraday prints of 35,500 on 2 October 2012, a loss of about 65 percent of value in twelve months and about 30 percent in the final week; trading was halted, and on reopening the rate recovered to about 28,500 (Observed, B). A subsidised official rate of 12,260 coexisted throughout: the gap, about 190 percent at the trough, was the tradable fact, and essentially no private actor could buy at the official rate.

The 2013 rally. This is the best-measured relief rally in the sample, and its size is routinely overstated in retelling. From about 36,200 before the June 2013 election, the rate reached about 29,900 within days of Rouhani's victory, an appreciation of 12–16 percent, then faded to about 32,000 by end-June (Observed, B). The Geneva interim agreement of 24 November 2013 took the dollar from about 30,000–31,000 to about 29,000, a further 3–4 percent on the day (Reuters, 2013); traders reported heavy state dollar supply, and the CBI was understood to be leaning against further strengthening. Cumulatively, by December 2013 the rial was about 20 percent stronger than immediately before the election (Iran Times, 2013). In the same month the CBI had already abolished the 12,260 subsidised rate and set the official reference at 24,779; from then on, unification pressure was resolved at every step by the official rate crawling toward the market rate, never the reverse.

2015–2016: the dog that did not bark. The Lausanne framework of 2 April 2015 produced no measurable move (33,300 the day before, 33,460 after) (Iran Times, 2015b). At the JCPOA signing on 14 July 2015 the free rate was about 32,000 (Associated Press, 2025); by Implementation Day, 16 January 2016, it was about 35,000–36,000, roughly 10 percent weaker than the June 2013 peak. The whole normalisation delivered stabilisation around 30,000–36,000 while Iranian consumer prices cumulated roughly 60–70 percent: a large real appreciation absorbed by domestic inflation, not nominal strength. Drift under relief followed: about 36,000 in mid-September 2017 and 41,981 on 31 December 2017 (RFE/RL, 2018).

April 2018: the failed unification. After a run to 60,000 on 9 April 2018, the government decreed a single rate of 42,000 for all purposes and declared trading at any other rate contraband; about 180 dealers were arrested and bureaus were banned from selling foreign exchange within days (RFE/RL, 2018). The arithmetic is the lesson: the official rate was devalued about 13 percent (37,000 to 42,000), and the market rate was ordered to appreciate about 30 percent by decree. Only the first half was within the government's power. Within a week the street rate was about 55,500; the 42,000 rate survived only as the subsidised essentials rate while trade migrated to the secondary system. Unification widened the effective multiple-rate structure it was meant to abolish.

2018–2020: collapse and partial retracement. Around the May 2018 US withdrawal the rate went from about 65,000 to 87,000 by end-June and 112,000 on 29 July (Associated Press, 2018), peaking at 190,000 on 26 September 2018 (Iran Times, 2018). Heavy intervention and exporter-repatriation rules then held a band of about 114,000–140,000 through much of 2019, still about 3.3× weaker than at the unification decree; the 2018 record was broken in June 2020 (204,800).

Did the 2013–2015 buyer make money? Table 4 reconstructs USD outcomes for a hypothetical holder who could buy and sell at the free rate with zero spread, a generous assumption.

Foreign money confirms the scale problem. The post-JCPOA vehicle of record, the Turquoise Partners and Charlemagne Capital Cyprus fund launched in January 2016, managed over 90 percent of all foreign portfolio investment on the Tehran Stock Exchange, which is itself the measure of how small the total was (Observed, B). By February 2020 the Securities and Exchange Organization estimated total foreign investment in listed companies at about USD 150 million at the market rate (Foundation for Defense of Democracies, 2020). Four years after relief became stated policy, foreign portfolio money in Iran's bourse was a rounding error, because bank de-risking kept payment and repatriation channels unreliable even in 2016–2018. Section 6.5 returns to these vehicles as alternatives.

4.2 The unification cases

Table 5 standardises the comparison set: six unifications or normalisations plus Iran's own 2018 decree. The regularity is the best-supported finding in the evidence base. In every case the official

Table 4: Iran: reconstructed USD returns to a frictionless free-market holder, 2012–2018

Entry	Exit	Holding period	USD return
2 Oct 2012 @ 35,500 (panic)	16 Jan 2016 @ ∼35,500	3.3 years	≈ 0%
13 Jun 2013 @ 36,200	23 Jun 2013 @ 29,900 (rally low)	∼10 days	+21%
23 Jun 2013 @ 29,900	31 Dec 2017 @ 41,981	4.5 years	−29%
14 Jul 2015 @ 32,000 (JCPOA)	9 Apr 2018 @ 60,000	2.7 years	−47%
14 Jul 2015 @ 32,000	26 Sep 2018 @ 190,000	3.2 years	−83%

Notes: (Reconstructed, B) from the prints sourced in this section. Three qualifications make the true picture worse than the table: offshore rial cash earns nothing (onshore deposits paid 15–22 percent but required surrendering dollars into the Iranian banking system); no legal, liquid offshore market in rial notes existed at scale, so realised spreads would have consumed much or all of the 2013 event gain; and every buy-and-hold window of a year or more that began in 2013–2015 lost money in USD, heavily if it ended after May 2018. The only reliably profitable version was a days-to-weeks event trade around the June and November 2013 headlines.

Source: Congressional Research Service (2014); Iran Times (2013); Reuters (2013); Associated Press (2025); RFE/RL (2018); Iran Times (2018).

rate moved to the market rate. The gap closed by official devaluation in seven of seven cases; the market rate appreciated to the official rate in zero of seven.

The two durable stabilisations in the set, Vietnam 1989 and Argentina 2023–25, share a structure the other five lack: monetary expansion actually stopped. Vietnam halted credit to the budget and raised deposit rates to positive real levels, and inflation collapsed within a quarter. Argentina combined the official devaluation with zero central-bank financing of the Treasury and dismantling of the quasi-fiscal stock. In the other cases, including Iran in every relief year, money growth stayed above 20 percent and no durable currency outcome of any kind followed. This makes the deciding variable of any rial scenario not the deal but the monetary leg: whether CBI financing of the deficit stops, observably, for six to eighteen months after the politics (Section 10.2 states this as a falsifiable condition).

Two side cases isolate mechanisms the main set cannot. Russia 2022 shows a quoted appreciation that no foreigner could realise: the ruble recovered to about 60 by June 2022 under a 20 percent policy rate, forced exporter FX sales, and a ban on foreign investors selling or withdrawing, then slid back through 2023 (Reuters, 2022). The controls that manufactured the rate froze the principal; an exchange rate under comprehensive controls is an administered number, and its appreciation is not evidence of a realisable opportunity. Iraq shows the retail base rate: the dinar was never revalued in two decades (it was devalued in 2020), the state regulator's standing alert records that dinars can be redeemed only in Iraq and that dealer markups ran to about 20 percent each way, and realised returns were negative even at a flat rate (Washington State Department of Financial Institutions, 2024). A depressed oil-state currency "must revalue" is a thesis with a documented base rate of about zero per decade, before frictions.

4.3 The patterns the model takes from the record

Five findings pass into Section 5 as inputs, each graded (Assumed, B) as judgement over the observed evidence. (a) Rally anatomy: credible normalisation news is worth 3–5 percent on a headline and 10–20 percent cumulatively over the best three-to-six-month window, front-loaded to the first credible headline and to the election or interim stage rather than to implementation; signing and implementation in Iran's case were worth about zero and negative drift respectively. No case in the set, including full normalisation (Vietnam) and full premium closure (Argentina 2024), produced a sustained appreciation above 25 percent in the rate a speculator can buy. (b) Unification is an

Table 5: Exchange-rate unification and normalisation episodes: how the gap closed, and what a street-rate holder received

Case	Gap before	How the gap closed	Outcome at the street rate
Myanmar 2012	Official 6.4 vs parallel ∼800–1,000	Official devalued ∼128× to a float at 818	Nothing at unification; kyat then fell ∼26% in three years
Vietnam 1989	Multiple rates, large parallel premium	Unified at the parallel rate (official devalued ∼5×), budget credit halted, real deposit rates positive	Dong depreciated with the inflation differential for 35 years
Sudan 2021	Official 55 vs parallel ∼375	Official moved to 375 under an IMF-monitored programme	Nothing at unification; closure held ∼8 months on donor FX, killed by the October 2021 coup
Cuba 2021	State 1:1, household 24:1	Single rate set at 24 (state sector devalued 24×)	Nothing realisable; a second official rate at 120 followed within 20 months
Argentina 2015	Official 9.9 vs blue ∼14.5	Controls lifted; official devalued to 13.95, onto the blue rate	Flat on day one; peso then depreciated through the normalisation
Argentina 2023–25	Premium ∼200% (blue >1,000 vs official 366)	Official devalued to 800, 2% crawl, zero central-bank financing of the Treasury	Premium closed to ∼1.4% by Dec 2024; blue-rate cash ∼1,000 was worth ∼1,050 a year later, about 0%
Iran 2018	Official ∼37,000 vs market 60,000	Decree at 42,000: official devalued, market ordered to appreciate	Decree failed within a week (street ∼55,500); 42,000 survived only as a subsidy rate

Notes: Grades: episode facts (OBSERVED, B) (Argentina 2023–25 premium path (RECONSTRUCTED, B); late Cuban informal prints (OBSERVED, C)). The pattern statement (official devaluation in 7 of 7) is a count over this table, not an estimate.

Source: Global Voices (2012); Reuters (2015b); International Monetary Fund (1993); International Monetary Fund (1995); Reuters (2021); The Sentry (2021); Reuters (2020); Hernández-Catá (2022); Cefeidas Group (2015); Nik Khosravi Pour (2025); RFE/RL (2018).

official-rate event (Table 5); a scenario that pays the holder requires appreciation of the free rate itself, a strictly stronger condition. (c) Openings release pent-up foreign-exchange demand faster than supply unless positive real rates or a fiscal anchor give residents a reason to hold the local currency; where the anchor preceded or accompanied liberalisation, the opening held. (d) Money growth is the discriminating variable: it stopped in exactly the two durable cases. (e) Relief consumes its credibility dividend within about two years when the monetary leg never arrives, which is why appreciation phases in the model are bounded at one to two years and followed by resumed drift in every relief scenario.

5 The scenario model

5.1 Construction, and the assumptions stated openly

The model prices one position: a foreign holder buys rials at the mid-October 2026 free-market rate and holds to six months, one, three, five, or ten years. The entry rate is $S_0 = 2{,}685{,}000$ rials per dollar, the midpoint of the documented range 2,670,000–2,700,000 (Observed, B). Five scenarios carry the probability weights of Section 4's judgement ranges, at their midpoints, summing to 1.0: status quo 0.500 (range 45–55 percent), partial relief 0.250 (20–30), comprehensive deal 0.125 (10–15), full reintegration 0.050 (3–7), instability 0.075 (5–10) (Assumed, B). Within each phase the rate compounds log-linearly, so each phase reproduces its stated total move exactly. Appreciation figures are gains in the rial's dollar value; the rate falls by the reciprocal.

The scenario paths rest on the following inputs. Every dynamic input is Assumed, because the precedent record supports ranges, not point paths; the ranges are Section 4's and the point choices are midpoints unless stated.

- **Status quo** (probability 0.500): blockade and sanctions persist or tighten; USD/IRR grows 32.5 percent per year throughout (band 25–40 percent, cross-checked against realised depreciation of 28–38 percent per year for 2018/19–2024/25). No relief foreign exchange arrives.
- **Partial relief** (0.250): waivers or tolerated oil sales, no banking reconnection. Rial value rises 17.5 percent over one year (band: +10 to +25 percent over 0.5–1.5 years), then USD/IRR grows 22.5 percent per year (band 15–30 percent). Relief foreign exchange at the low case of Table 3, about USD 26.5 billion per year, is the plausible scale.
- **Comprehensive deal** (0.125): a verified nuclear deal with oil, banking, and SWIFT relief, defined with the Central Bank tightening condition holding. Rial value rises 30 percent over two years (band: +20 to +40 percent over one to three years, stated in the source as an outer bound conditional on tightening), then USD/IRR grows 15 percent per year. The 15 percent post-phase drift is a model choice, not a source figure: it is set at the bottom of the partial-relief resumed-drift band to reflect the scenario's tightening condition, and it flatters the scenario. Relief foreign exchange at the base case, about USD 51.5 billion per year, plus the one-off storage sale of USD 5–8 billion.
- **Full reintegration** (0.050): deal plus unification and monetary reform on the Vietnam or Argentina-II sequence. The premium closes mainly by official-rate devaluation (seven of seven precedents), so the free rate gains little: value rises 15 percent over four years (band: 0 to +30 percent over three to five years), then USD/IRR grows 10 percent per year. The 10 percent drift is a model choice anchored to the Vietnam template, where the nominal rate depreciated

roughly with the inflation differential after 1989. Relief foreign exchange at the high case, about USD 79.5 billion per year, is attainable only here.

- **Instability** (0.075): political transition or escalation shock. Rial value falls 50 percent inside the first quarter (band: 40–60 percent in weeks, on the October 2012 and September 2018 episodes), then status-quo drift resumes from the shocked level. No retracement is modelled, although the 2012 and 2018 episodes retraced part of the spike within a year; the omission is conservative for the holder at short horizons.

Two treatments are computed throughout. The **theoretical** treatment liquidates at the modelled Tehran free-market rate with zero frictions and cash held at 0 percent nominal; it is an upper bound that no offshore holder can realise, and it is labelled as such wherever it appears. The **realisable** treatment, computed in Section 7, runs the same paths through the exit evidence of Section 6. The model's further Assumed inputs (informal entry and exit haircuts, the deposit schedule's extension beyond its documented years, the reading of the account turnover cap as a position cap, and the redenomination window close) are stated in Section 7.1 where they bind, and collected in the limitations, Section 11. Sanity checks run in code on every execution: probabilities sum to 1.0 and sit inside the source ranges; every path matches an independent closed-form recomputation of its segment assumptions; no scenario pays a frictionless holder 1.5× at any horizon; realisable recovery never exceeds the same holding form's free-rate value.

5.2 Paths and theoretical returns

Table 6 gives the scenario value multiples of a cash position, and Figure 1 plots the rate paths.

Table 6: Scenario definitions and USD value multiples of a frictionless cash position

Scenario	Prob.	6m	1y	3y	5y	10y
Status quo	0.500	0.869	0.755	0.430	0.245	0.060
Partial relief	0.250	1.084	1.175	0.783	0.522	0.189
Comprehensive deal	0.125	1.068	1.140	1.130	0.855	0.425
Full reintegration	0.050	1.018	1.036	1.111	1.045	0.649
Instability	0.075	0.434	0.377	0.215	0.122	0.030

Notes: Multiples are S_0/S_h for the scenario rate S_h at each horizon: the USD value of a cash position per USD 1 entered, before all frictions (MODELLED, B). Probabilities are the midpoints of the judgement ranges in Section 4 (ASSUMED, B). The best single outcome in the grid is partial relief at one year, 1.175×. No scenario pays more than +18 percent at any horizon, consistent with the precedent finding that no case pays an IRR holder more than +50 percent in USD in any scenario.

Source: desk model, model.py and results.json (data appendix).

The shape of the result is visible in the table. Relief gains are front-loaded and then eroded: partial relief peaks at one year and is below entry value by year three; even the comprehensive path, with its outer-bound appreciation and a deliberately flattering post-phase drift, is below entry value by year five. Only full reintegration holds a gain at five years (1.045), and it carries a probability of 0.050. The status quo, at half the weight, compounds at 32.5 percent per year against the holder in every period.

Table 7 is the paper's main results table: probability-weighted theoretical outcomes. Returns are identical across stakes in this treatment (there are no fixed costs and no caps), so the stakes differ only in dollar values.

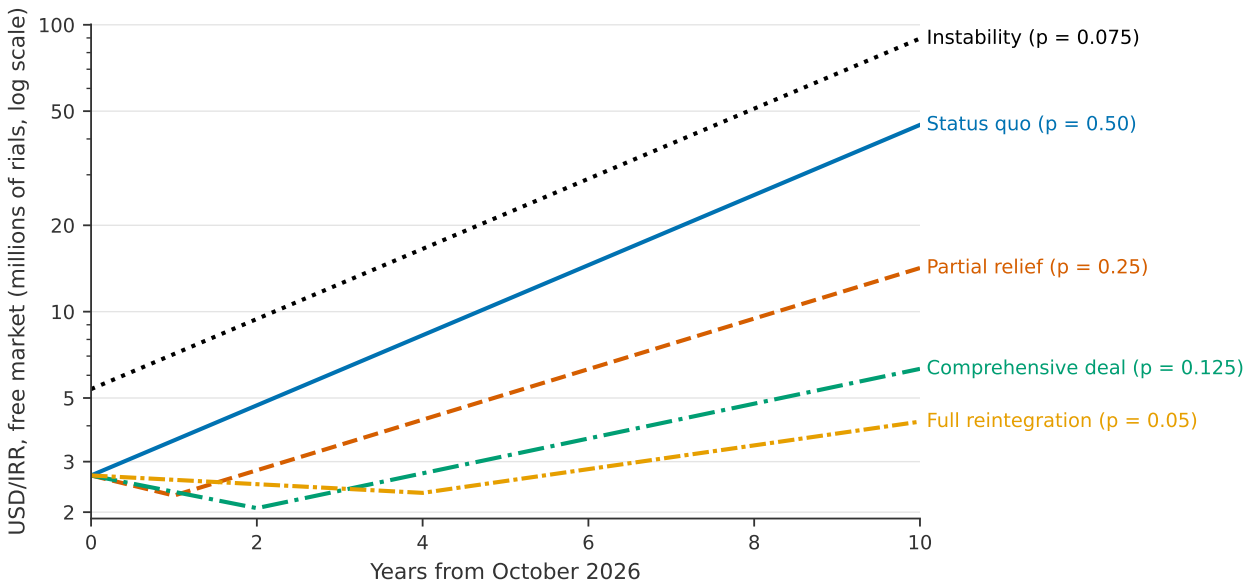


Figure 1: Scenario paths for the Tehran free-market USD/IRR rate

Source: desk model (model.py), computed with the model's own path function from $S_0 = 2{,}685{,}000$. Note: log scale. The instability path applies the 50 percent value shock inside the first quarter and therefore leaves the entry level immediately. Relief paths dip below the entry rate during their appreciation phases and then rise as resumed depreciation dominates; every path ends above the entry rate, including full reintegration.

Table 7: Main results: probability-weighted theoretical values and returns (frictionless upper bound)

Horizon	Multiple	Return	Value of $1,000	Value of $10,000	Value of $100,000
6 months	0.922	−7.8%	$922	$9,223	$92,230
1 year	0.894	−10.6%	$894	$8,937	$89,371
3 years	0.624	−37.6%	$624	$6,236	$62,364
5 years	0.421	−57.9%	$421	$4,212	$42,118
10 years	0.165	−83.5%	$165	$1,651	$16,509

Notes: (MODELLED, B) outputs of the model in Section 5.1, probability-weighted over the five scenarios with a single weight set at all horizons (a limitation: Section 11, item 1). "Theoretical" means liquidation at the modelled Tehran free-market rate with zero frictions. At the documented Western bid of USD 0.00 (Section 6.4), none of this value is recoverable outside Iran; Section 7 computes what is.

Source: desk model, results.json.

5.3 Why the weighted result is negative at a tenth of PPP

The result surprises only if the PPP gap is treated as a return forecast. It is not one, for three reasons the evidence makes concrete. First, returns are path objects, and the modal path is continued depreciation: a 32.5 percent annual depreciation at probability one half outweighs bounded appreciation phases of +17.5 to +30 percent in total, at combined probability 0.425, because the appreciation is a level shift and the depreciation is a compounding rate. Partial relief's entire gain is +17.5 percent once; the status quo takes 24.5 percent in the first year alone and keeps taking it. Second, the relief appreciation is bounded by the precedent record, not by the PPP gap: the largest sustained appreciation in any scenario or precedent is +40 percent in total (the comprehensive outer bound), and no unification in the set moved the market rate toward the official rate at all. A convergence of the free rate toward PPP, or even toward the official complex, would be an event outside the observed set, and Section 8 quantifies how far outside. Third, the gap itself partly prices the impairments of Section 2.4 (blocked exports, fiscal dominance, controls), which the relief scenarios only partly remove: even the comprehensive path assumes banking relief and tightening, yet resumes 15 percent annual depreciation because nothing in the evidence supports assuming Iranian money growth falls to single digits, a fall that has occurred in no year in the record, relief years included. The undervaluation is the value of an option whose payoff the precedent record caps at about +20 to +40 percent, struck against a modal outcome of continued 25–40 percent annual depreciation. Priced that way, the option is out of the money in expectation.

6 Executability

This section documents, for the investor profile of Section 1, how a rial position would be acquired, held, and exited, and what each step costs or forbids. The finding is stated first: the trade fails on executability before it reaches valuation. Entry is possible only informally or at trivial size; holding is carry-negative in every lawful form; exit has no lawful offshore bid.

6.1 The legal bars

United States, primary sanctions. The Iranian Transactions and Sanctions Regulations (31 CFR 560) block the property of the Government of Iran and of Iranian financial institutions, including the Central Bank of Iran, whenever it is in the United States or held by a US person. Rial balances are liabilities of the CBI and of Iranian banks, so a rial position that touches US jurisdiction engages the blocking regime directly. Section 560.208 separately prohibits facilitation by US persons of transactions by non-US persons that would be prohibited for US persons: any US bank, US broker, US-resident intermediary, or US-dollar clearing leg in the chain converts a non-US person's purchase into a US-jurisdiction event (Observed, A). Enforcement is current and aimed at the plumbing this trade needs: on 15 October 2026 OFAC's Operation Economic Outcast designated 16 individuals and several entities, including the International Development Investment and Finance Bank of Iraq, Iran's Chabahar International Offshore Bank and Jazireh International Bank, and several currency exchange companies, principally under Executive Order 13902 (U.S. Department of the Treasury, 2026; Mondaq, 2026) (Observed, B).

The rial-specific provision. Executive Order 13645 of 3 June 2013 authorises sanctions on foreign financial institutions that knowingly conduct or facilitate significant transactions for the purchase or sale of the Iranian rial, or that maintain significant accounts outside Iran denominated in the rial. The stated objective at issuance was to render the currency unusable outside Iran. The order was

never rescinded; its JCPOA-era suspension ended with the 2018 withdrawal, and it remains in force in October 2026 (The White House, 2013) (OBSERVED, A). The sanction falls on intermediaries, not on the retail buyer: no designation of a retail currency buyer was found in the record (RECONSTRUCTED, B). Its effect is nonetheless decisive for this trade, because a bank or licensed exchanger that intermediates rial transactions at scale risks its US correspondent access, so regulated intermediaries in third countries do not openly run a rial book. The provision works by removing the market-maker. Section 6.4 documents the resulting absence of a bid.

UN, EU, and UK measures. The E3 triggered the JCPOA snapback on 28 August 2025, and UN Security Council resolutions 1696, 1737, 1747, 1803, 1835, and 1929 were reimposed effective 27 September 2025 (China, Russia, and Iran dispute the legality; the measures are applied by the EU, the UK, and allied states) (OBSERVED, B). Council Regulation (EU) 2025/1975 reinstated the asset freeze on the CBI and major Iranian banks, the SWIFT ban, and prohibitions on transfers between EU institutions and Iranian financial institutions or bureaux de change (Council of the European Union, 2025; Harneys, 2025). The UK designated 71 persons and entities including major financial institutions (OBSERVED, B). For this investor the EU and UK regimes matter less as personal prohibitions than as a further removal of intermediation: after September 2025 no EU or UK regulated institution can lawfully settle a rial transaction chain involving Iranian financial institutions.

Canada. Canada's regime under the Special Economic Measures (Iran) Regulations is a listings-based dealings prohibition, not a comprehensive embargo: purchasing Iranian currency as such is not an offence under Canadian Iran sanctions, and the prohibition attaches to designated persons and their property (Baker McKenzie, 2025) (OBSERVED, B). The practical exposure differs from the formal position. The realistic counterparties in this trade (Iranian exchange houses, sarrafi networks, offshore Iranian banks) are heavily designated under the US and EU regimes, so a purchase from a designated counterparty becomes a prohibited dealing; and no Canadian bank will process the payment leg, because the bank's own US and EU exposure governs its conduct. The Global Affairs Canada sanctions page could not be loaded from this network during the research; the Canada summary rests on the law-firm record and search extracts and is graded accordingly.

No relief is in force. The 17 June 2026 Islamabad memorandum of understanding, which set a framework for ending the 2026 conflict, collapsed within days. Iran's September 2026 UN General Assembly proposal (reopen Hormuz, nuclear talks, in exchange for lifting the blockade and oil sanctions and releasing frozen assets) was publicly rejected by President Trump. Reuters reported on 7 October 2026, citing an Iranian official, that no nuclear negotiations were taking place (OBSERVED, B). As of the writing date there is no live, scheduled relief event that a position could be carried toward: any entry today is a carry-negative, exit-blocked position held against an unscheduled political event.

6.2 Acquisition

The offshore cash market closed at the top end in 2012, when Reuters reported that major UAE exchange houses, including Al Ansari Exchange, had stopped dealing in the rial under devaluation and US regulatory pressure (Reuters, 2012) (OBSERVED, B). Since then the organised end of the channel has been progressively designated (Section 6.1). A licensed exchanger in Dubai, Istanbul, Erbil, Kabul, or Yerevan openly quoting a two-way rial price to a walk-in foreign retail buyer was not found in this research; published bid–ask spreads for physical rials from licensed third-country dealers are Unknown, and the absence is recorded as a research result, not a gap in effort. What trades is the expatriate and trade-settlement sarrafi network: bilateral, relationship-based, priced by negotiation against the Tehran free rate, with the offshore leg normally settled in dirhams, lira, or USDT rather than banknotes. It is also the network OFAC designates. A foreign retail speculator

has no documented access point into it that does not run through an unlicensed intermediary.

The one openly accessible offshore market in rial notes is the numismatic trade, and its prices prove it is not a currency market: a dealer set of 12 uncirculated notes (100 to 1,000,000 rials) sells for USD 29.99, while the face value of the largest note in the set is about USD 0.37 at the October 2026 free rate (Observed, B). Buying through this channel pays a large multiple of face for an asset whose monetary bid is zero (Section 6.4).

Crypto routes run the wrong way: they are the exit technology Iranian residents use, not an entry for foreigners, and the channel is a designated battlespace. Tether froze about USD 550 million of Iran-linked USDT in 2026, of which USD 344 million in April alone (Observed, C); OFAC has designated the Iranian exchanges Nobitex and Wallex among others, and Nobitex was hacked in June 2025 with roughly USD 90 million destroyed (CoinTurk, 2026) (Observed, B). A USDT claim on a designated Iranian exchange is unsecured exposure to that counterparty, denominated in a token whose issuer freezes Iran-linked wallets; it captures no rial appreciation in enforceable form.

Inside Iran, the only formal purchase channel open to a visiting foreigner is the CBI's tourist scheme of October 2025: an authorised bank buys the tourist's hard currency and issues a rial debit card, converting at the rate discovered in the Center for Exchange of Currency and Gold. On the scheme's first reported day the conversion rate was 1.01 million against a free-market rate of 1.07 million, a haircut of about 5.6 percent to the free market before any spread on exit (Press TV, 2025a) (Observed, C). The scheme is built for spending, not accumulation. Street purchase at the free rate from licensed Tehran sarrafi is physically straightforward for a person present with cash; it converts the problem rather than solving it, since the buyer then holds notes inside a capital-controlled border (Section 6.3) or a deposit inside the capped account regime (Section 6.3), and the exit problem of Section 6.4 applies in full.

Size is a physical fact before it is a financial one. At 2.675 million per dollar, a USD 10,000 position is 26.75 billion rials: 26,750 notes in the largest commonly circulating denomination (the IRR 1,000,000 note or CBI cheque), or 267,500 notes in 100,000-rial banknotes; a USD 100,000 position is ten times that (Reconstructed, C). Accumulation at investment size is a logistics and security exercise in bricks of paper, in jurisdictions where the cargo has no lawful market, with no offshore authentication infrastructure and no recourse against an informal seller.

6.3 Holding

A CBI regulation permits non-resident foreign individuals to open rial savings accounts and term deposits, in person, at a credit institution, on presentation of a passport. The same regulation caps the annual turnover of such accounts at IRR 5,000,000,000 per credit institution, subject to triennial indexation (Central Bank of Iran, 2023) (Observed, A). At the model's entry rate the cap is USD 1,862 per year per institution. Whether the cap as currently applied differs from the hosted text is Unknown; no updated figure was located. Balances so held sit in a banking system severed from SWIFT, with the CBI and major banks under EU asset freeze, and with no deposit-insurance equivalence a foreigner could realistically claim against.

Carry is negative in every lawful form. The CBI's published deposit schedule (Money and Credit Council circular of 30 January 2023, shown unchanged for 1401–1403) sets one-year term deposits at 20.5 percent, two-year at 21.5 percent, three-year at 22.5 percent (Observed, A); cash earns 0 percent (Observed, A). Against inflation above 35 percent in late 2025 the best lawful deposit carry was negative in real terms by roughly 12 to 20 percentage points a year; against the SCI twelve-month average of 73.6 percent to September 2026, the one-year rate is a real return of about minus 31

percent (Reconstructed, B). A rial position must therefore appreciate at a double-digit real rate before the holder breaks even in purchasing-power terms. Section 7 models the deposit only up to the capped slice, because the cap binds: the slice is 100 percent of a USD 1,000 stake, 18.6 percent of USD 10,000, and 1.9 percent of USD 100,000.

The border rule completes the holding regime. A Money and Credit Council regulation permits travellers, Iranian or alien, to import or export at most IRR 500,000 in rial notes; amounts beyond that are dealt with under Article 42 of the Monetary and Banking Law (Central Bank of Iran, 2020) (Observed, A). At the entry rate the lawful export quantity is about USD 0.19. No superseding figure was located. Foreign cash is separately controlled: CBI rules require documentation to hold or carry foreign banknotes above EUR 10,000 equivalent, with undocumented excess to be deposited or sold to credit institutions (Press TV, 2025b) (Observed, B). The round trip therefore fails at both ends: the rials cannot lawfully leave in size, and the hard currency cannot lawfully move undocumented in size.

6.4 Exit

Exit is the binding constraint, and it is documented price by price.

Outside Iran, there is no bid. A major US-based foreign-currency buyback dealer lists the IRR 1,000,000 note with a buyback value of USD 0.00 (Foreign Currency and Coin, 2026) (Observed, B). Mainstream travel-money bureaux do not quote the rial at all. The informal sarrafi bid is the only offshore bid of any size, and it is a negotiated relationship price for settlement balances, not a posted price for banknotes; a quantified discount to the Tehran free rate for physical notes held abroad is Unknown, because no verifiable quote, listing, or transaction report was located. The structural evidence (EO 13645's design since 2013, the 2012 Dubai withdrawal, the zero buyback quote) indicates the discount is severe and the market thin to the point of absence for a seller without an existing network. Modelling offshore exit as "free-market rate minus a spread" is therefore not a defensible assumption; Section 7 models it as an explicit set of cases, including zero.

Inside Iran, exit requires presence and documentation. Licensed exchangers will sell hard currency against rials to a person physically present, at or near the free rate, for amounts consistent with the foreign-cash documentation rules. There is no remote, banked, or brokered exit for a non-resident retail holder. Exit also degrades exactly when it would be attempted: in every documented Iranian foreign-exchange crisis the state responded to scarcity by tightening allocation, widening the gap between administered and free rates, and policing unofficial dealing (Reconstructed, B).

The official channel does not cover this trade. FIPPA guarantees licensed foreign investors repatriation of capital and returns, but conversion is at the rate "as acknowledged by the Central Bank", transfers require three months' prior notice, Foreign Investment Board approval, ministerial confirmation, and prior payment of taxes, and permits issue case by case (Government of Iran, 2002) (Observed, A). A currency purchase is not a licensable FIPPA investment, so a retail rial holder has no FIPPA protection at all; and even covered investors have historically seen local gains converted into hard-currency losses at administered rates (Section 4.1).

The outcome record. No documented case was found of a foreign retail holder of physical rials realising a profit by converting back to hard currency (Reconstructed, B). Individual-scale anecdotes of travellers arbitraging small sums inside Iran exist; none is documented at investment scale, and none survives the IRR 500,000 border cap as a repeatable strategy.

7 Realisable returns

7.1 Construction

The realisable treatment runs the Section 5 paths through the Section 6 evidence. Entry is charged at one of three haircuts: the documented tourist-card case, 5.6 percent of rials not received (OBSERVED, C); and two Assumed informal cases, 10 percent (central) and 25 percent (adverse), recorded as scenario haircuts rather than measurements because the informal sarrafi discount is unquantified in any verifiable source. The position splits into a deposit slice of at most IRR 5 billion (the account cap of Section 6.3, read here as a cap on the deposit placed through one institution; the reading is Assumed, the cap itself (OBSERVED, A)) and a cash slice for the remainder. The deposit slice earns 20.5 percent nominal per year, the documented one-year schedule (OBSERVED, A) rolled at all horizons; the extrapolation beyond the schedule's documented years is Assumed and flagged in Section 11. Cash earns 0 percent. Exit takes one of four cases:

- **X0, documented bid.** The deposit slice recovers zero (no documented repatriation channel for a retail currency holder; FIPPA does not cover currency purchases). The cash slice recovers only the border-cap quantity, IRR 500,000, at the free rate.
- **X1, informal bid.** The whole position recovers its free-rate value less an Assumed haircut of 25 or 50 percent. No verifiable source quantifies the informal bid for physical notes; the qualitative anchor is the documented retail round trip of up to about 20 percent each way in the Iraq dinar analogue (Section 4.2). These cases are Assumed throughout and are labelled as such in every table.
- **X2, in-person Tehran exit.** The holder is present in Iran, sells at the free rate, and carries out at most USD 11,600 in cash, the documented EUR 10,000 foreign-cash threshold converted at the EUR/USD cross of 1.16 implied by the free-market rates (RECONSTRUCTED, B); any excess recovers zero. No travel cost is charged, because no cost figure exists in the evidence base. This is an Assumed upper variant, not a documented channel at size.
- **Redenomination hazard (all cases).** The conversion window is modelled to close at $t = 4.0$ years (October 2030), from the statute's outer timeline (enacted October 2025, up to two years of preparation, then a three-year dual-circulation window) (MODELLED, B). If the horizon exceeds the close, the cash slice recovers zero in every case, including X2: unconverted old notes held offshore have no conversion route and a documented bid of zero. The deposit slice is not exposed, because bank balances convert automatically at the statutory ratio. The plausible close range is roughly 2029 to 2031; the five- and ten-year cash outcomes are zero under any close date before those horizons.

7.2 Results

Table 8 gives probability-weighted realisable returns under the central entry case (10 percent Assumed haircut), and Figure 2 plots the USD 10,000 stake against the theoretical bound.

Three structural facts drive the numbers.

First, the documented bid is zero, and the arithmetic is exact about it. Under X0, recovery on a USD 10,000 stake at one year is between USD 0.07 (instability) and USD 0.22 (partial relief) depending on the scenario, USD 0.17 probability-weighted: the border cap of IRR 500,000 is the only lawful quantity of notes that can leave, and everything else has a bid of zero. For a USD 1,000

Table 8: Realisable returns, probability-weighted (percent; central entry case: 10 percent Assumed informal entry haircut)

Stake	Exit case	1 year	3 years	5 years
$1,000	X0 documented bid	−100.0	−100.0	−100.0
	X1 informal, 25% haircut	−27.3	−26.3	−27.8
	X1 informal, 50% haircut	−51.5	−50.9	−51.8
	X2 in-person Tehran	−3.1	−1.8	−3.7
$10,000	X0 documented bid	−100.0	−100.0	−100.0
	X1 informal, 25% haircut	−37.1	−51.4	−85.1
	X1 informal, 50% haircut	−58.1	−67.6	−90.0
	X2 in-person Tehran	−16.2	−35.4	−80.1
$100,000	X0 documented bid	−100.0	−100.0	−100.0
	X1 informal, 25% haircut	−39.4	−57.3	−98.5
	X1 informal, 50% haircut	−59.6	−71.5	−99.0
	X2 in-person Tehran	−88.4	−88.4	−98.0

Notes: (Modelled, B) paths run through the Section 6 exit evidence; the X1 haircuts and the 10 percent entry haircut are Assumed, the X0 bid and the border cap are (Observed, A/B), and the X2 cap is (Reconstructed, B). Under the documented entry haircut (5.6 percent) the figures improve modestly (for example the USD 10,000 X2 one-year return is −12.2 percent rather than −16.2 percent); X0 is unchanged. The full grid, all entry cases, all scenarios, and the six-month and ten-year horizons are in the model output (data appendix).

Source: desk model, results.json.

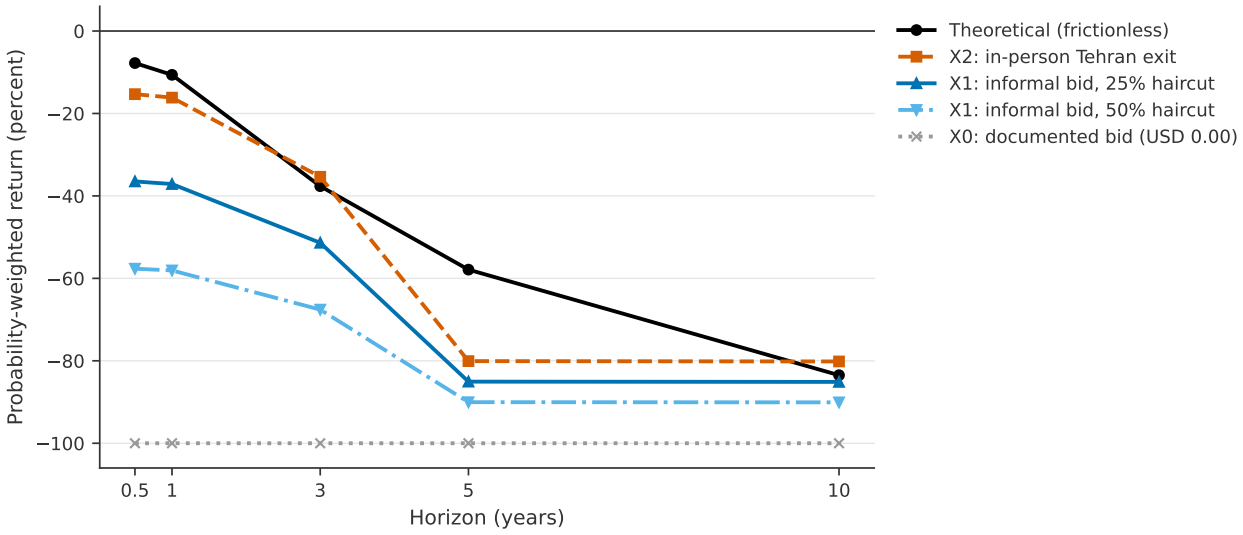


Figure 2: Theoretical and realisable probability-weighted returns, USD 10,000 stake

Source: desk model, results.json (central entry case for the realisable lines). Note: the theoretical line is the frictionless upper bound of Table 7; X0 is flat at −100 percent because recovery is between USD 0.07 and USD 0.22 at every horizon shown. The drop in the X1 and X2 lines between three and five years is the modelled redenomination window close (October 2030), after which the cash slice recovers zero and only the capped deposit slice survives.

stake the recovery is exactly zero, because the whole stake sits inside the deposit slice, which has no documented repatriation channel. The return is minus 100.0 percent at every horizon and every stake, for practical purposes exactly.

Second, scale hurts. The in-person case caps recovery at USD 11,600 of proceeds, so a USD 100,000 stake loses 88.4 percent at one and three years even where the rate cooperates, while a USD 1,000 stake, which fits entirely inside the deposit cap and earns 20.5 percent nominal on the whole position, roughly breaks even under X2 and under the favourable informal case. The USD 10,000 stake sits between: mostly cash, fully exposed to the exit wall and, past year four, to the redenomination hazard.

Third, the redenomination hazard is the cliff at five years. At the USD 10,000 and USD 100,000 stakes the position is mostly cash notes. Held offshore past the modelled window close, the cash slice recovers zero in every exit case, and only the capped deposit slice (USD 1,862 of principal at entry) survives. That is why the five-year realisable returns collapse relative to three years in X1 and X2, while the theoretical line (which assumes conversion is always possible at the free rate) declines smoothly.

One corner case is reported rather than suppressed. For the USD 1,000 stake under full reintegration at ten years, realisable recovery under the Assumed exit cases (USD 2,828 at a 25 percent informal haircut; USD 3,771 in person) exceeds the theoretical cash benchmark (USD 649). The comparison is between holding forms, not between exit channels: the theoretical benchmark holds cash at 0 percent, while a USD 1,000 position fits entirely inside the account cap and is modelled earning 20.5 percent nominal for a decade under an appreciating-then-stable rate. The result depends on extrapolating the 2023 deposit schedule for ten years and on Assumed exit channels that Section 6 does not document at any size. It does not generalise to larger stakes, where the deposit slice is 18.6 percent of the position or less. The deposit slice is in any case bounded: at USD 1,862 of principal it cannot carry a position of USD 10,000 or more, and if Iranian deposit rates were cut or the account regime tightened, the slice would do worse than modelled.

8 The multiples test

The pitch is rarely made in percentages; it is made in multiples. This section states what 2×, 5×, and 10× require, and compares the requirement with the documented envelope.

A frictionless holder multiplies by k if the exit rate falls to S_0/k: 1,342,500 rials per dollar for 2×, 537,000 for 5×, 268,500 for 10×. Note what these levels are: the 5× exit rate is approximately the free-market rate of September 2024, and the 10× exit rate is approximately the World Bank's PPP factor rolled forward (Section 2.4). The multiples pitch is, in levels, a claim that the rate returns to where it stood two years ago (5×) or converges to purchasing power parity (10×). Table 9 converts the requirement into the sustained annual appreciation of the rial's dollar value that each multiple needs at each horizon.

Against the precedent record the verdict is uniform. No precedent in the comparison set pays a rial holder more than +50 percent in USD in any scenario; the two successful stabilisations produced street-rate convergence gains of about 0 percent (Vietnam over 35 years of nominal depreciation; Argentina-II blue cash about flat over the premium-closure year). A 2× outcome at any horizon requires sustained appreciation of a kind absent from the set; 5× and 10× require it for longer. These are not scenario outcomes. They would have to be trading outcomes, which is the remaining route to test.

Table 9: Sustained annual appreciation of the rial's USD value required for each multiple (percent per year)

Multiple	6 months	1 year	3 years	5 years	10 years
2×	+300	+100	+26.0	+14.9	+7.2
5×	+2,400	+400	+71.0	+38.0	+17.5
10×	+9,900	+900	+115.4	+58.5	+25.9

Notes: Required rate is $k^{1/h} - 1$ for horizon h in years, before all frictions (MODELLED, B). Comparison envelope from Section 4: the largest sustained appreciation in any scenario or precedent is +40 percent in total (the comprehensive-deal outer bound over one to three years); the best documented episode is about +20 percent over six months (2013) before retracement; the model's own best scenario pays 1.175×. Every cell exceeds the documented envelope, most by an order of magnitude.

Source: desk model, results.json.

The entry-timing route, and why it fails. The multiples could in principle be earned by entering at a panic print rather than at the October 2026 rate. Holding the scenario paths fixed, a 2× under partial relief at one year would have required entry at 4,570,213 rials per dollar, and under the comprehensive path at three years entry at 4,750,385: buying into a panic print 70 to 77 percent weaker than the mid-October 2026 rate, and then capturing a relief appreciation that itself sits at the top of the precedent range (MODELLED, B). The documented panic prints do not offer that combination to a buyer. The deepest spike-to-stabilisation moves in the record (September 2018: a 190,000 spike against a 2019 band of about 114,000–140,000; October 2012: 35,500 against a 28,500 reopening) are worth at most about 1.3 to 1.5× to a perfectly timed entrant, before frictions, with perfect foresight about which print is the spike. And the offshore holder cannot convert even that into dollars at size: the Western bid is zero and the lawful export quantity is IRR 500,000 (Section 6.4). A strategy that buys after a comprehensive deal is signed fares no better on this evidence: the rally is front-loaded to the first credible headline (2013: +20 percent before any sanctions were lifted; 2015 signing: about 0 percent; 2016 implementation: negative drift), so the post-signature buyer purchases after the appreciation has happened and been retraced once.

9 Alternatives

If the thesis is "Iran normalises and Iranian assets re-rate", the rial is not the only expression. This section examines the three alternatives the evidence base covers and reaches the same wall by different doors.

Tehran Stock Exchange equities through foreign-investor structures. The channel exists in Iranian law and is used at trivial scale. Securities and Exchange Organization data for late October 2025 put total foreign portfolio holdings at IRR 211.1 trillion (about USD 211 million at the then free-market rate) across 5,182 active foreign investors, of whom 1,770 were individuals; a large share is attributed by Iranian officials to Iranian expatriates (Tehran Times, 2025) (OBSERVED, B). The 2015–2017 vehicles show the operating constraints. Turquoise Partners managed over 90 percent of foreign portfolio investment; its Cyprus fund with Charlemagne Capital (January 2016) and Sturgeon Capital's 2015 fund were the structures that worked, and their documentation barred investors with any US link (finews, 2016; Reuters, 2015a) (OBSERVED, B), which excludes the investor profile of this paper. After the 2018 withdrawal the vehicles wound down, froze, or shrank under the repatriation mechanics of Section 6.4; their current status is Unknown in detail, and none was found open to a Canadian retail investor today. On payoff structure, TSE equities are a rial asset whose nominal index tends to rise with inflation and devaluation: the foreign holder gains rial exposure plus equity

risk, then loses the currency gain at the administered-rate exit. It is the rial trade with an equity overlay and the same exit wall.

Iranian sovereign Eurobonds. None are outstanding. Iran has issued internationally twice in the post-1979 era, two euro-denominated issues totalling about EUR 1 billion in 2002, both long matured (International Monetary Fund, 2006) (OBSERVED, A); a USD 500 million issue announced in December 2015 for the post-JCPOA reopening never came to market (Iran Times, 2015a). There is no Iranian sovereign Eurobond to buy at any price in October 2026, and the EU's reinstated measures also prohibit transactions in Iranian government securities.

Oil instruments. Brent, WTI, oil equities, and oil-linked currencies are legally trivial for this investor to buy and capture the scenario's oil-price leg. They contain no rial payoff: they pay on the commodity path whether or not the rial appreciates, and oil strength under continuing sanctions has historically coexisted with rial collapse (the October 2026 record lows were printed alongside a blockade, not an oil shortage in Iran's favour). As a proxy for the rial thesis, oil is a hedge against one scenario driver, not an expression of the thesis.

The summary is exact: no legal instrument available to this investor pays off if and only if the rial appreciates. The thesis is, today, untradeable through regulated markets.

10 Verdict and falsifiability

10.1 Verdict

The trade fails twice, independently. It fails on expected value: with zero frictions, at Tehran prices, on scenario weights and appreciation bounds taken from the most favourable reading of the precedent record, the probability-weighted return is negative at every horizon (-10.6 percent at one year, -37.6 at three, -57.9 at five, -83.5 at ten) (MODELLED, B), and the best single scenario pays $1.175\times$. It fails on executability: at the documented Western bid of USD 0.00, the realisable recovery on a USD 10,000 position is USD 0.07 to USD 0.22 at one year, and every formal channel in the chain is size-capped between USD 0.19 (the border quantity of notes) and USD 1,862 (the annual account turnover cap). The undervaluation that motivates the pitch is real (Section 2.4) and irrelevant to the outcome, because the two mechanisms that could convert undervaluation into a holder's return, relief-driven appreciation of the free rate and unification, are both contradicted by the precedent record in the form the pitch requires: appreciation is bounded at +20 to +40 percent in total and retraces, and unification has closed the gap by official devaluation in seven of seven cases. The multiples the pitch trades on ($2\times$, $5\times$, $10\times$) require sustained nominal appreciation with no precedent in the set, including in the two successful stabilisations.

For completeness, the verdict is profile-specific. A holder inside Iran, able to hold deposits at 20.5 percent, sell at the free rate in person, and consume in rials, faces a different and less unfavourable calculation, though still a negative real carry at current inflation. A holder with a pre-existing sarrafi relationship faces the Assumed informal cases of Section 7 rather than the documented bid; those cases still lose money at every stake of USD 10,000 and above at the horizons that matter, and they rest on channels that are designated, unlicensed, and unquantified. Neither qualification rescues the trade for the investor profile this paper addresses.

10.2 Falsifiability

The conclusions rest on documented prices, statutes, and the precedent record, and they are falsifiable by observable changes. Each of the following, if it occurred and were documented at the standard of evidence used in this paper, would change the answer in the stated direction.

1. **A licensed offshore market with a real bid.** If licensed dealers in any major third-country centre resume quoting two-way prices in physical rials (as Dubai did before 2012), with a published bid within, say, 25 percent of the Tehran free rate, the X0 exit case is wrong and realisable returns should be recomputed at that bid. This is the single change that would most alter Section 7.

2. **Positive real deposit rates.** If the CBI's deposit schedule rises above the SCI inflation rate on a sustained basis, the holding regime stops being carry-negative and the deposit-slice analysis, including the corner case of Section 7.2, generalises upward in stake size. The test is mechanical: the published schedule against the published SCI twelve-month rate.

3. **Demonstrated monetary restraint after a deal.** If, following any relief agreement, Iranian broad-money growth falls below about 15 percent annualised and stays there for six to eighteen months (the window the precedent record associates with durable stabilisation), the post-relief drift assumptions in the relief scenarios are too high and the appreciation phases should be extended. The deal itself, without this observation, changes nothing in the model: the 2013–2016 episode is the evidence.

4. **An exchange provision for offshore notes in the redenomination.** If the CBI's implementation rules create a channel for holders outside Iran to convert old notes (through correspondent banks, embassies, or licensed exchangers) within the dual-circulation window, the redenomination hazard of Section 7.1 should be removed or repriced, and the five- and ten-year realisable returns rise accordingly.

5. **A documented profitable exit at size.** A single verified case of a foreign retail holder converting a rial position of USD 10,000 or more back to hard currency at within 25 percent of the free-market rate, through any channel, would contradict the exit finding of Section 6.4 and require its revision. None was found; the finding is an absence, and absences are falsifiable by production of the instance.

11 Limitations and robustness

1. **One probability set for all horizons.** The scenario weights are judgements about the dominant regime over a five-year horizon. Applied at six months and one year they overstate the weight on relief at short horizons: as of writing there were no nuclear talks in being (Section 6.1), so a relief path dominating within six months is less likely than the flat weights imply. Short-horizon weighted results should be read with that bias in mind; per-scenario results are unaffected.

2. **Log-linear phases.** Real relief rallies are front-loaded to the first credible headline, not spread evenly over the phase (Section 4.3). The convention is neutral on totals and optimistic on timing risk: realised short-horizon outcomes would be more timing-sensitive than the model shows.

3. **Assumed post-phase drifts.** The comprehensive deal's 15 percent and reintegration's 10 percent post-phase depreciation rates are model choices without a source figure. Both are below every Iranian historical outcome in the evidence base, so the relief scenarios are, if anything, flattered at long horizons.

4. **The informal haircuts are Assumed, not measured.** The 10 and 25 percent entry haircuts and the 25 and 50 percent exit haircuts have no verifiable source, because the informal market publishes nothing. They are presented as cases, and the documented-bid case (X0), which requires no assumption about informal prices, is the headline. If the true informal discount is smaller than 25 percent, the X1 cases improve toward the theoretical line; they do not cross it, and they remain negative at USD 10,000 and above on the modelled paths.

5. **Deposit schedule extrapolation.** The 20.5 percent one-year rate is documented for 2023 and shown unchanged through 1403 (March 2024–March 2025). The model rolls it for up to ten years. The exposure is bounded by the cap (USD 1,862 of principal), so the error this can introduce is small at stakes of USD 10,000 and above, and material only in the USD 1,000 corner case of Section 7.2, which is flagged there.

6. **The account cap is a turnover cap in the source regulation.** The model reads it as a position cap through one institution. A holder using several institutions could place more in deposit form. The currently indexed figure is Unknown. The reading affects only the deposit slice, which is bounded as in item 5.

7. **The in-person exit case is generous.** X2 assumes the holder can travel, sell at the free rate, and carry out USD 11,600, with no travel cost (no cost figure exists in the evidence base) and no tightening of controls at the moment of exit. Section 6.4 records that controls tighten exactly then, and that exit value in a crisis is unproven. X2 is an Assumed upper variant, not a documented channel at size.

8. **The redenomination close is modelled, not scheduled.** The close at $t = 4.0$ years follows the statute's outer timeline; the plausible range is roughly 2029 to 2031. If preparation runs shorter, the cliff arrives before the three-year horizon instead of after it. Secondary reporting also conflicts on the new unit's name; the mechanics are unaffected.

9. **Entry dispersion.** Trackers diverge by 5–10 percent in fast markets and day-to-day moves reach about ±5 percent (Reconstructed, B). All results scale with the entry print actually obtained; the model uses the documented range midpoint.

10. **Tehran prints.** The paths are Tehran free-market rates. Regional and informal rates differ; the model prices that dispersion only through the entry and exit haircut cases.

11. **Stream B magnitudes are plausibility anchors only.** No elasticity from oil foreign exchange to the exchange rate is estimated, because the evidence base provides none. The relief scenarios' exchange-rate paths come from the precedent record (Section 4), not from the oil arithmetic (Section 3); a reader who believes the flow effect is larger than any precedent delivered must supply the mechanism the record lacks.

12. **Legal-change risk is carried qualitatively.** Nothing in the model prices the probability that the position becomes legally problematic for the holder mid-holding, through designation of the channels themselves (Section 6.1 documents the direction of travel). That risk is real, documented, and unquantifiable from the evidence base; it is an overlay on the numbers, all of which are computed as if the holder's legal position were static.

13. **Data vintage and the two inflation series.** Market and macro data are as of 16 October 2026. Iran's two official inflation series disagree by about 5 percentage points; the paper reports both and uses neither as a model input beyond the real-carry comparison. The IMF's 2026 output projection differs by vintage (-5.4 vs -6.1 percent); the paper flags the spread where the figure appears.

12 Conclusion

A foreign retail investor cannot buy the Iranian rial in any sense that survives contact with the evidence. The currency is genuinely cheap on price levels, and that fact does no work: relief has never paid a holder of a currency like this one more than a bounded, front-loaded, retracing appreciation; unification has never paid the street-rate holder anything; the scenario arithmetic on precedent-based weights loses money before frictions; and the friction at the end of the chain is total, a documented bid of zero behind a border cap of nineteen cents, inside a legal architecture designed in 2013 to produce exactly that result. The correct statement of the opportunity is narrower than the pitch and is worth keeping, because conditions change: the rial is an option on Iranian monetary stabilisation, currently untradeable offshore, whose payoff conditions are observable and are listed in Section 10.2. If a licensed bid reappears, if deposit rates turn positive in real terms, or if money growth falls and stays down after a deal, this paper's answer should be revisited. On the evidence of October 2026, the answer is no.

Acknowledgments and production note

This paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data. The scenario model (model.py, with its documentation and full output) and the four research streams underlying Sections 2, 3, 4, and 6 are retained by the desk and summarised in the data appendix. This document is research, not investment advice and not legal advice.

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A Data appendix

A.1 Series definitions and sources

Table 10: Series definitions and sources

Series	Definition and source	Coverage / vintage	Grade
Free-market USD/IRR	Cash dollar, Tehran open market; Bonbast-tracked prints via WSNext and Reuters	Point prints, Sep–Oct 2026	Observed, B
ICE negotiated rate; SANA reference	CBI exchange-centre negotiated hall (sell) and CBI published average of licensed offices; via Etemad Online and CBI daily release	Point prints, Oct 2026	Observed, B
Official rate (legacy)	WDI PA.NUS.FCRF; frozen at 42,000, 2019–2023, series ends 2023	To 2023	Observed, A
PPP conversion factor	WDI PA.NUS.PPP, GDP basis	2025: 147,938	Observed, A
Atlas conversion factor	WDI PA.NUS.ATLS	2025: 742,604	Observed, A
Inflation	SCI CPI (12-month average and point-to-point) and CBI CPI; IMF WEO PCPIPCH	To Sep 2026	Observed, B/C/A
Money and credit	CBI broad money and liquidity, via CEIC and CBI Economic Trends	To Sep 2025 (levels); 1405 growth statements	Observed, A/B
Output and fiscal	IMF WEO NGDPD, NGDP_RPCH, GGXCNL_NGDP, GGXWDG_NGDP	2026 values are projections	Observed/Modelled, A
Production	OPEC Monthly Oil Market Report, secondary sources, Table 5-7	To Sep 2026	Observed, A/B
Loadings	Kpler and Vortexa tanker tracking, via Reuters	To Oct 2026	Observed, B
Reserves and frozen assets	No official series exists; estimates per Section 2.3	Apr 2026 breakdown	Assumed, C
Deposit schedule	CBI Money and Credit Council circular, 30 Jan 2023, in CBI Economic Trends No. 121	Shown unchanged 1401–1403	Observed, A
Scenario paths and returns	Desk model (model.py); inputs and grades in Sections 5.1 and 7.1	Horizons 0.5–10 years	Modelled, B

A.2 Frozen and blocked assets: the April 2026 breakdown

Table 11: Iranian frozen and blocked assets by location, April 2026 estimates

Location	Estimate ($bn)	Note
China	∼20	Oil proceeds in restricted arrangements; other reports 20–50
India	∼7	Oil receivables
Iraq	∼6	Electricity and gas receivables, largely restricted-use
South Korea → Qatar	∼6	2023 transfer, still restricted-use
United States	∼2	Including litigated Clearstream-held funds
Japan	∼1.5	
EU / Luxembourg	∼1.6	
Memo: Iranian official claims of inaccessible assets	>100	
Memo: accessible reserves (unattributed)	∼24	
Memo: realistically releasable on a deal	∼30–35	Much pre-committed

Notes: All entries (Assumed, C): indicative, not audited. The location breakdown is from widely cited April 2026 explainers; the releasable subset is from 2023-era independent analyses. Iran publishes no reserves total; the total is Unknown (Section 2.3). The CBI separately acquired at least USD 507 million of USDT during 2025 to support the rial, of which Tether froze about USD 344 million in April 2026 (CryptoBriefing, 2026) (Observed, C).
Source: Khabarwalay (2026); Factually (2026); CryptoBriefing (2026).

A.3 Model reproducibility

The model is a single pure-Python file (model.py) with a separated inputs block in which every input carries its stream source and evidence grade. It writes the full output grid (results.json): scenario paths, value multiples, theoretical values and returns for all stakes and horizons, and realisable recoveries and returns for all combinations of three entry cases, four exit cases, three stakes, five scenarios plus probability-weighted, and five horizons. Sanity checks (probability sum and range membership; closed-form path recomputation; monotonicity of the status-quo path; the 1.5× frictionless bound; realisable never exceeding the same-form free-rate value; weighted-multiple recomputation) run on every execution and passed at the vintage used for this paper (October 2026). Tables 6, 7, 8, and 9 and Figures 1 and 2 are drawn from that output without hand-editing.